Tutorial 3: Building a Neural Network for MNIST Handwritten Digit Classification

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1. Introduction

This report details the development of a neural network to classify handwritten digits (0–9) using the MNIST dataset. The project focused on comparing architectures, observing model performance across different complexities, and implementing training safeguards like Early Stopping.

2. Methodology

2.1 Dataset and Preprocessing

The MNIST dataset was loaded and normalized. The training data was split into a 48,000-image training set and a 12,000-image validation set to monitor for overfitting.

```python
# Split training into train and validation
train_size = int(0.8 * len(train_dataset))
val_size = len(train_dataset) - train_size

train_dataset, val_dataset = random_split(train_dataset, [train_size, val_size])

train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=32)
test_loader = DataLoader(test_dataset, batch_size=32)
```

Python

```
100%|██████████| 9.91M/9.91M [00:00<00:00, 18.6MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 488kB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 3.91MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 5.86MB/s]
```

2.2 Model Architectures

Three distinct models were implemented to compare performance:

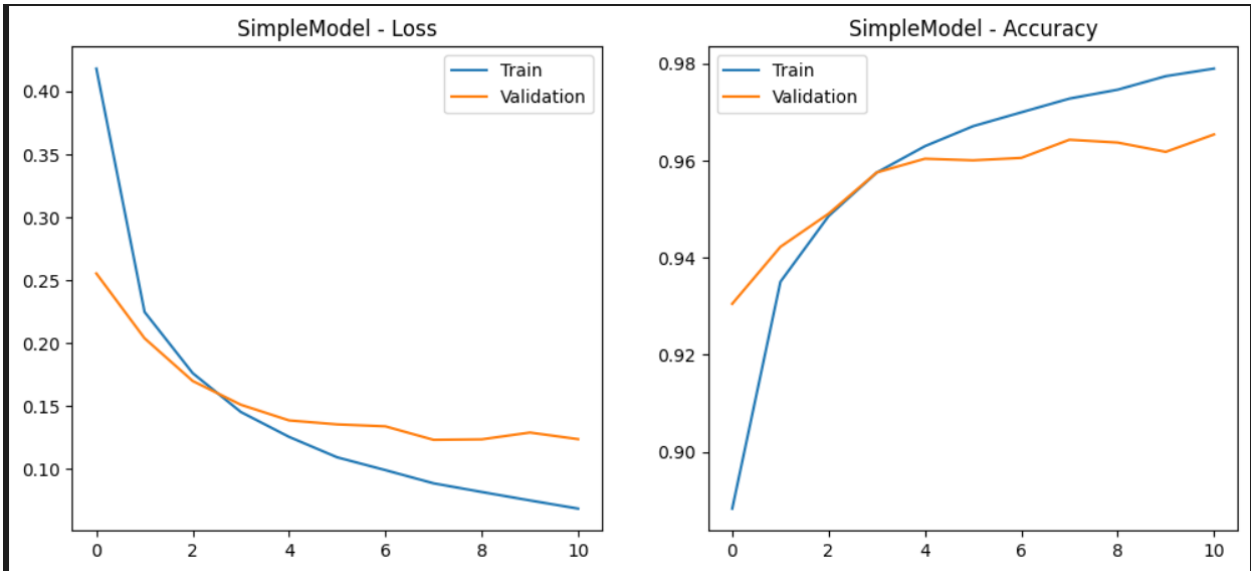
1. **Simple Model:** 1 hidden layer (32 units).
2. **Base Model:** 2 hidden layers (128 and 64 units).
3. **Complex Model:** 3 hidden layers (256, 128, and 64 units).

3. Experimental Results

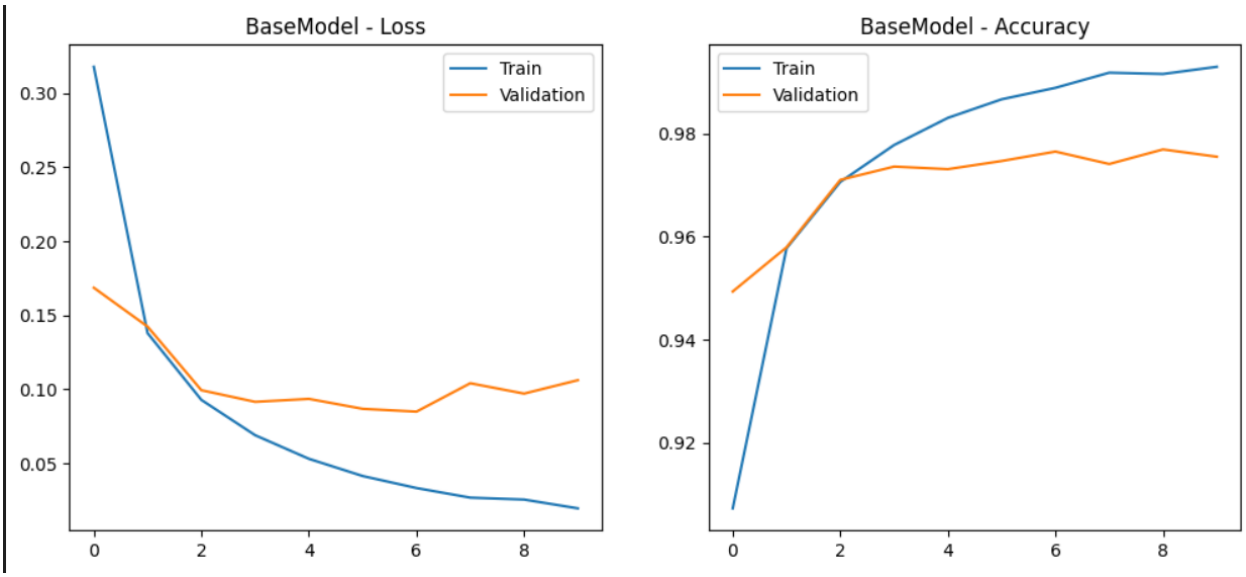
3.1 Training Analysis (Loss & Accuracy Curves)

The following plots illustrate how each model learned over 20 epochs.

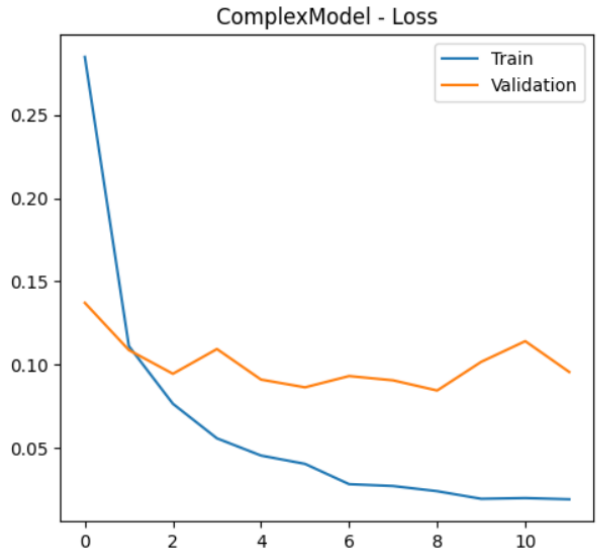
A. Simple Model

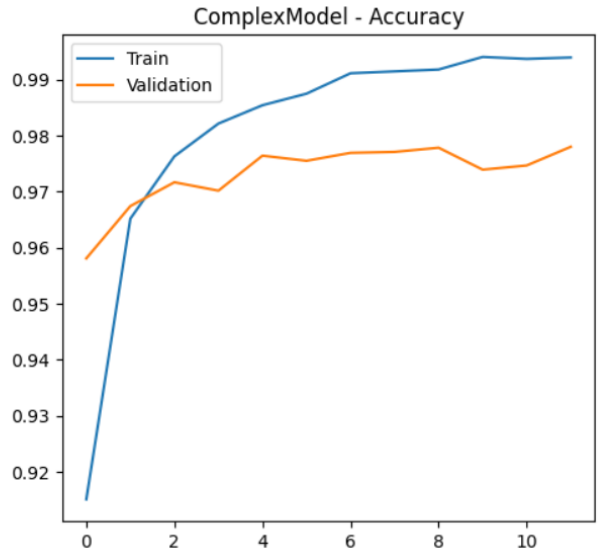


B. Base Model



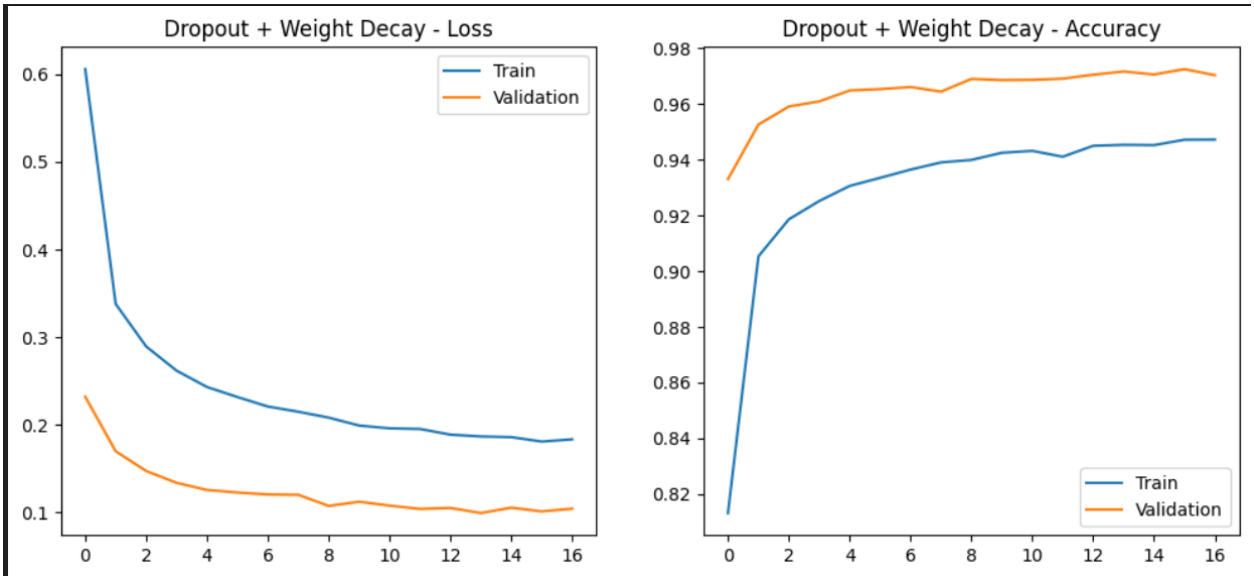
C. Complex Model





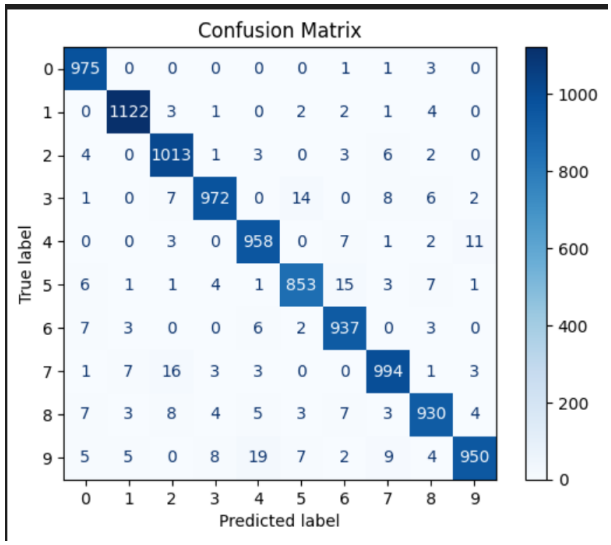
3.2 Final Model Evaluation

The best-performing model was evaluated on the held-out test set to determine its real-world accuracy.



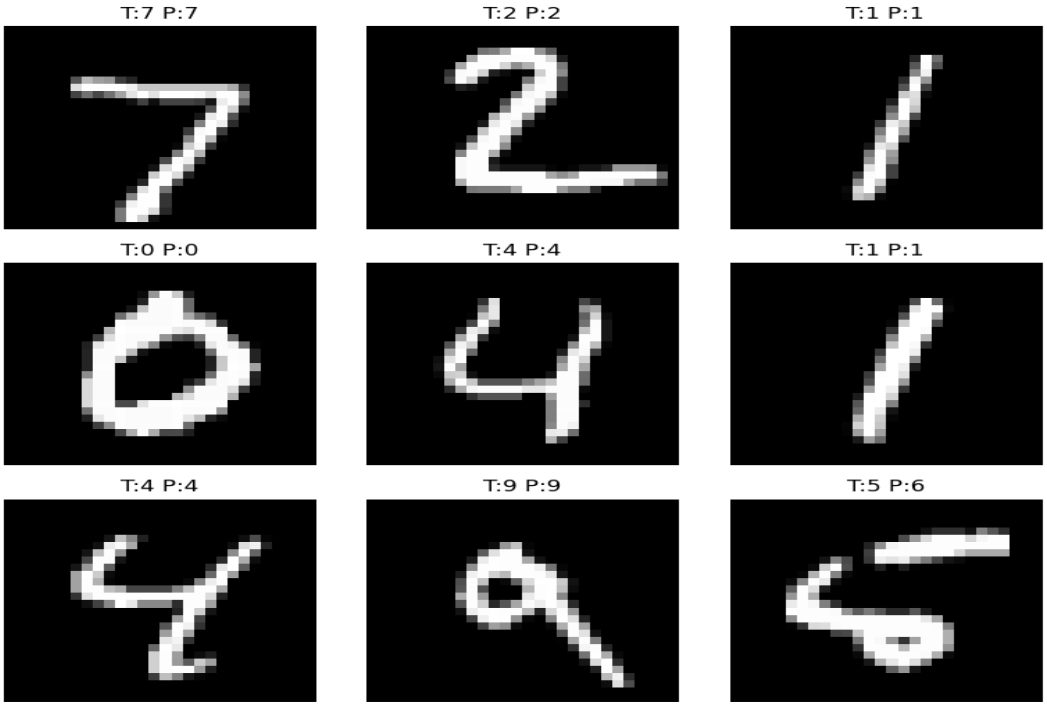
Confusion Matrix

To analyze specific misclassifications, a confusion matrix was generated.



3.3 Prediction Validation

To verify the model's practical performance, a subset of the test images was passed through the network to compare the predicted labels against the ground truth.



- **Observation:** The model correctly identifies most digits, including those with slight variations in handwriting style. Any misclassifications here usually involve digits that are visually ambiguous even to the human eye.

4. Discussion and Conclusion

- **Model Comparison:** The Simple Model showed signs of underfitting, while the Complex Model showed a wider gap between training and validation performance, indicating potential overfitting.
- **Efficiency:** The **Base Model** provided the most stable results.
- **Early Stopping:** This technique proved vital in preventing the models from training longer than necessary once the validation loss stopped improving, saving computational time and ensuring better generalization.